

Software Requirements Specification for

Event Booking System, Release 1.0

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
MennaTallah Ahmed	15 / 5 / 2025	Initial Draft	1.0 Draft 1
Nour Mohammed	17 / 5 / 2025	Changes Made to (2. Overall Description)	1.0 Approved

1. Introduction

1.1 Purpose (of the Document):

This document provides a detailed Software Requirements Specification (SRS) for the Event Booking System. It is intended for stakeholders, developers, testers, and project managers to understand the goals, features, constraints, and interactions of the system. The document serves as a baseline for system design and development. It describes the functional and non-functional requirements for release 1.0 of the Event Booking System (EBS).

1.2 Project Scope and Product Features:

The Event Booking System will be a mobile and web-based platform with a friendly [UI] that allows users to browse, book, and manage event tickets seamlessly. Users (attendees) can search for events, choose seats, and pay through multiple secure payment gateways, receiving a QR-coded e-ticket upon confirmation. Event organizers can create and manage events, track ticket sales, and validate attendance. Admin users can monitor transactions, manage users and events, and generate platform-wide reports.

Features covered in this release include user registration and login, event browsing and search, real-time seat selection, ticket purchasing, [QR-code] generation, and dashboard functionalities for users, organizers, and administrators.

The system supports real-time operations, ensures secure transactions [PCI-DSS], ensures users' data security [GDPR] and provides analytics and reporting for organizers and admins.

1.3 Definitions, Acronyms, and Abbreviations:

- EBS: Event Booking System
- UI: User Interface
- MFA: Multi-Factor Authentication
- QR Code: Quick Response Code
- PCI-DSS: Payment Card Industry Data Security Standard
- GDPR: General Data Protection Regulation

1.4 References:

1. Event Booking System Vision and Scope Document, Event Booking Team – Phase 1
2. Event Booking System Requirements Elicitation Document, Event Booking Team – Phase 2
3. Google Forms Survey and Interview Notes, Event Booking Team, April 2025
4. World Wide Web Consortium (W3C). Web Content Accessibility Guidelines (WCAG) 2.1. <https://www.w3.org/WAI/WCAG21/quickref/>
5. PCI Security Standards Council. PCI DSS Quick Reference Guide, v4.0. <https://www.pcisecuritystandards.org>

General Data Protection Regulation (GDPR) Compliance Overview. European Commission. https://commission.europa.eu/law/law-topic/data-protection_en

2. Overall Description

2.1 Product Perspective

The Event Booking System is a new cross-platform application designed to replace traditional manual, phone, and third-party services for managing event registrations and ticketing. The system will serve as an intermediary between event organizers and attendees, facilitating event creation, promotion, and booking.

This system is expected to evolve over several phases, ultimately integrating with external calendar tools, payment gateways, and social media platforms. It will operate as both a mobile and web application with a shared backend and centralized database for consistency.

A context diagram for release 1.0 will illustrate major actors such as attendees, organizers, payment services, and notification services.

2.2 User Classes and Characteristics

- **Event Attendee (Primary User) :**
An Attendee is a registered user who uses the system to browse events, select seats, book tickets, make payments, and receive QR-coded digital tickets. Attendees may also review events, receive notifications, and manage bookings.
 - Estimated user base: ~10,000 across both platforms.
 - Most will access the app via smartphones, some through the web.
 - Expected frequency: 2–4 bookings/month.
 - Users may want to receive reminders or cancel bookings depending on event policies.
- **Event Organizer :**
Organizers use the system to create and manage events, sell tickets, apply discount codes, track attendee stats, and validate QR tickets at the venue.
 - Can be individuals, companies, or agencies.
 - Responsible for refund policies and event cancellations.
 - Must verify identity before gaining full access to organizer tools.
- **System Administrator :**
Admins manage platform content, moderate users and events, and oversee financial operations.
 - Limited to a few trusted system operators.
 - Will resolve disputes, detect fraud, and approve or reject events before publishing.
- **Payment Gateway Service :**
An external system used for processing credit card, debit card, and digital wallet transactions securely.

2.3 Operating Environment

- **OE-1 :** The Event Booking System shall support the following web browsers:
 - Google Chrome (latest 2 versions)
 - Mozilla Firefox (latest 2 versions)
 - Safari (latest version for macOS and iOS)
 - Microsoft Edge (latest version)
- **OE-2 :** The server-side application shall operate on the corporate-approved versions of Red Hat Enterprise Linux with Apache Web Server or Nginx.

- OE-3 : The system shall be accessible from the corporate Intranet and from external Internet connections, subject to security policies and authenticated access through a secure login.
- OE-4 : Payment services shall be integrated through secure APIs

2.4 User Documentation

2.4.1 Introduction

The Event Booking System is a platform designed for users to discover, book, and manage event tickets seamlessly. It supports:

- Attendees (users booking events)
- Event Organizers (managing events & ticket sales)
- Administrators (managing platform operations)

UD-1 : An online help center will guide users through browsing, booking, and payment processes.

UD-2 : A first-time user tutorial will be shown to attendees and organizers during onboarding, including a walkthrough of ticket booking and event creation (respectively).

UD-3 : The system will offer FAQs, troubleshooting tips, and customer support contact through the help section.

2.4.2 Attendee Guide

1.Registration & Login

How to Register:

1. Click "Register" on the homepage.
2. Choose registration method:
 - Email & Password
 - Phone Number
 - Social Login (Google/Facebook)

3. Fill in required details (name, contact info).
4. Confirm via email/SMS (if applicable).

How to Login:

1. Click "Login" on the homepage.
2. Enter your email/phone & password or use social login.
3. Click "Submit" to access your account.
- ☐ Forgot Password? Click "Reset Password" and follow the email instructions.

2. Event Browsing & Booking

1. Browsing Events:

- After logging in, go to the Home or Explore page.
- Use filters to search by:
 - Event type (concert, workshop, conference, etc.)
 - Date
 - Location
 - Price
- Click on any event card to view:
 - Event details
 - Venue
 - Date & Time
 - Organizer info
- Available ticket types

2. Booking Tickets:

- On the event details page, select:
 - Ticket type (General Admission, VIP)
 - Quantity
- Click "Book Now" or "Add to Cart"
- Proceed to Checkout
- Enter payment details (credit/debit card, digital wallet, etc.)
- Confirm the booking
 - A confirmation email/SMS will be sent with the e-ticket

3. Ticket Management

1. Viewing Booked Tickets:

- Go to My Bookings in your profile menu
- View:
 - Upcoming events
 - Past events
 - QR codes or ticket numbers for entry

2. Canceling or Transferring Tickets:

- Select a booking
- If cancellation is allowed, click "Cancel Booking" and confirm
- For eligible events, use "Transfer Ticket" to send the ticket to another registered user via email

4. Profile & Preferences

1. Editing Profile:

- Go to My Profile
- Update name, contact info, profile picture
- Save changes

2. Notification Preferences:

- Choose how to receive updates:
 - Email
 - SMS
 - In-app notifications
- Manage preferences in Settings > Notifications

5. Support & Help

- For help, click "Support" in the footer or profile menu
- Access:
 - FAQs
 - Live Chat
 - Submit a support request
- For urgent issues, contact the hotline (if available)

2.5 Design and Implementation Constraints

- CO-1 : The system shall be developed using modern web technologies (e.g., React or Angular for frontend, Node.js or Django for backend).
- CO-2 : Database operations must conform to ACID properties using a relational DBMS such as PostgreSQL.
- CO-3 : QR code generation shall follow standard formats compatible with third-party scanners.
- CO-4 : All communications must be encrypted via HTTPS.
- CO-5 : Integration with payment gateways must comply with PCI-DSS.

2.6 Assumptions and Dependencies

- AS-1 : Internet access is required for all system operations.
 - AS-2 : Event organizers are responsible for refund and cancellation policies, not the platform itself.
 - DE-1 : The system depends on third-party APIs for payment processing, SMS/email delivery, and QR code scanning.
 - DE-2 : Future expansion depends on integration with social media platforms and external calendar systems.
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3. System Features

3.1 Functional Requirements

3.1.1 User Registration (FR-01)

Description:

Allows new users to create accounts via email, phone, or social login

Elicitation Technique:

- Interview (with Mr. Ayman Salah, IT Manager)
- Survey (87.5% of respondents wanted easy registration)

How Applied:

- Interview revealed need for multiple registration options

- Survey confirmed social login as preferred method for 50% of users

3.1.2 Login (FR-02)

Description:

Authenticates users via credentials or social accounts. Includes password recovery.

Elicitation Technique:

- Observation (noted manual password resets in existing systems)
- Prototype Feedback (Nour El-Din requested biometric login)

How Applied:

- Observation of paper-based recovery → implemented email/SMS reset flow
- Prototype suggestion → added fingerprint/FaceID for mobile

3.1.3 Event Booking and Search (FR-03)

Description:

Enables users to book tickets with seat selection and payment

Elicitation Technique:

- Observation (noted issues with real-time availability)
- Prototype feedback (Ahmed Hamed requested ticket limits)

How Applied:

- Observation showed need for real-time seat updates (<2s delay)
- Prototype feedback led to "ticket limit per user" feature

3.1.4 Seat Selection (FR-04)

Description:

Interactive seat map for reserved-seating events with real-time availability.

Elicitation Technique:

- Interview (Mr. Ayman Salah: "Option to choose my seat is critical")
- Observation (noted 15-20 min wait times for manual seat assignment)

How Applied:

- Interview → color-coded seat map with tooltips
- Observation → implemented 120-second seat locking during booking

3.1.5 Ticket Booking (FR-05)

Description:

End-to-end booking flow with payment processing.

Elicitation Technique:

- Survey (87.5% prioritized "booking confirmation" feature)
- Prototype Feedback (Mohamed Ali requested fewer steps)

How Applied:

- Survey → added instant QR ticket generation
- Prototype → reduced steps from 7 to 5 with progress indicator

3.1.6 Payment Processing (FR-06)

Description:

Supports credit cards, PayPal, and digital wallets.

Elicitation Technique:

- Interview (Ms. Hagar Ehab recommended Stripe/Razorpay)
- Survey (50% preferred digital wallets)

How Applied:

- Interview → integrated Stripe API with PCI-DSS compliance
- Survey → added Apple Pay/Google Pay options

3.1.7 E-Ticket Management (FR-07)

Description:

Generates QR-coded tickets with download/email options.

Elicitation Technique:

- Observation (paper sign-ins in Entrepreneurship Course → QR need)
- Prototype Feedback (Ahmed Hamed: "QR scan must work offline")

How Applied:

- Observation → implemented offline-valid QR codes
- Prototype → added "Save to Wallet" button

3.1.8 Event Creation (FR-08)

Description:

Organizers can create events with tickets, pricing, and seating charts.

Elicitation Technique:

- Prototype Feedback (Ahmed Hamed requested "Duplicate Event" feature)
- Interview (Ms. Hagar Ehab stressed need for bulk upload)

How Applied:

- Prototype → added "Copy Event" button with editable templates
- Interview → implemented CSV import for ticket types

3.1.9 Attendee Check in (FR-09)

Description:

QR scanning at venues with real-time validation.

Elicitation Technique:

- Observation 2 (inconsistent attendance tracking → standardized QR)
- Survey (66.7% valued "quick entry" at events)

How Applied:

- Observation → designed scanner with audio/visual confirmation
- Survey → added batch scan mode for groups

3.1.10 Cancellation and Refund (FR-10)

Description:

Handles user cancellations per organizer policies.

Elicitation Technique:

- Survey (87.5% wanted "cancel/edit bookings")
- Document Analysis (noted refund disputes in Observation 1)

How Applied:

- Survey → implemented self-service cancellation portal
- Document analysis → added automated refund status emails

3.1.11 Notification (FR-11)**Description:**

Sends booking confirmations, reminders, and alerts.

Elicitation Technique:

- Survey (54.2% wanted event reminders)
- Observation 2 (no-shows → automatic ban policy)

How Applied:

- Survey → scheduled SMS/email reminders (48h/24h before event)
- Observation → integrated no-show detection system

3.1.12 Reporting (FR-12)**Description:**

Generates sales, attendance, and revenue reports for organizers.

Elicitation Technique:

- Prototype Feedback (Sara Khaled requested monthly reports)
- Interview (Ms. Hagar Ehab recommended data exports)

How Applied:

- Prototype → added filters by date/event type
- Interview → enabled PDF/Excel export

3.1.13 Admin Dashboard (FR-13)**Description:**

Allows admins to manage users, events, and platform settings.

Elicitation Technique:

- Prototype Feedback (Sara Khaled needed role-based access)
- Interview (Mr. Ayman Salah emphasized audit logs)

How Applied:

- Prototype → implemented RBAC (Admin/Organizer/User tiers)
- Interview → added timestamped action logs

3.1.14 Social Sharing (FR-14)**Description:**

Users can share events on social media.

Elicitation Technique:

- Survey (33.3% wanted social integration)
- Prototype Feedback (Nour El-Din suggested wishlist)

How Applied:

- Survey → added share buttons for Facebook/Twitter
- Prototype → implemented "Save for Later" feature

3.1.15 Review and Rating (FR-15)**Description:**

Users can rate events (1-5 stars) and leave comments.

Elicitation Technique:

- Survey (open-ended responses highlighted review importance)
- Observation 1 (lack of feedback in club selection process)

How Applied:

- Survey → implemented moderated reviews with like/dislike
- Observation → added organizer response capability

3.2 Software System Attributes

ID	Category	Importance (1-5)	Requirement	Measurement Criteria
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NFR-01	Performance	5	System response time under normal load	System response time under normal load ≤2 seconds for page loads, ≤3 seconds for search results
NFR-02	Performance	5	Concurrent user handling	Support 5,000 simultaneous users during peak sales
NFR-03	Availability	5	System uptime	99.9% availability (≤8.76h downtime/year)
NFR-04	Reliability	5	Transaction integrity	0.01% maximum failure rate for confirmed bookings
NFR-05	Security	5	Data protection	AES-256 encryption at rest, TLS 1.3 in transit
NFR-06	Security	4	Authentication	Multi-factor authentication (MFA) for organizers/admins
NFR-07	Usability	4	Task completion time	≤2 minutes for ticket booking (5-step max workflow)
NFR-08	Usability	3	Accessibility	WCAG 2.1 AA compliance (e.g., screen reader support, color contrast ≥4.5:1)
NFR-09	Scalability	5	Vertical scaling	Handle 3x traffic surge during event launches without performance degradation
NFR-10	Maintainability	3	Code modularity	80%+ unit test coverage, documented APIs
NFR-11	Portability	4	Cross-platform support	Android (API 28+), iOS (13+), Web (Chrome/Firefox/Safari/Edge latest 2 versions)
NFR-12	Localization	3	Language/currency support	English/Arabic languages, USD/EGP currencies
NFR-13	Interoperability	3	Calendar integration	Sync with Google Calendar/Outlook via standardized APIs
NFR-14	Recovery	4	Data restoration	≤4 hours recovery time from backups (daily encrypted backups)
NFR-15	Compliance	4	Legal regulations	GDPR compliance for EU users, local data protection laws
NFR-16	Auditability	4	Activity logging	100% logging of financial transactions and admin actions, retained for 365 days
NFR-17	Fault Tolerance	4	Payment failure handling	Automatic seat release within 10 minutes of failed payments

NFR-18	Capacity	3	Database storage	Support 100,000+ events and 1M+ users
NFR-19	Latency	5	Real-time updates	≤1 second delay for seat availability changes
NFR-20	Notification Speed	4	Confirmation delivery	≤30 seconds for booking confirmations (email/SMS)

4. External Interfaces

4.1 User Interfaces

The system will include the following user interfaces:

- Home Page: Highlights events and categories & displays featured events and categories.
- Login/Registration Page: Secure access for users and organizers
- Event Listings: Allows users to filter events and search by category, location, date, price
- Event Details Page: Descriptions, timings, venue, price, booking options
- Seat Selection: Real-time seat layout with availability status
- Payment page: Secure interface for entering payment details and confirming bookings.
- User Dashboard: Enables users to view and manage bookings, download tickets, and cancellations
- Organizer Dashboard: Event creation, ticket management, reports
- Admin Panel: Offers moderation, analytics, and fraud detection functionalities
- [All UIs will be responsive and accessible on desktop and mobile platforms.]

4.2 Hardware Interfaces

The system does not require special hardware beyond standard computing devices (PCs, tablets, smartphones). QR code validation can be performed using mobile devices or barcode scanners compatible with standard USB input or mobile apps.

4.3 Software Interfaces

The system will integrate with:

- Authentication APIs: Google, Facebook, and phone/email login providers.

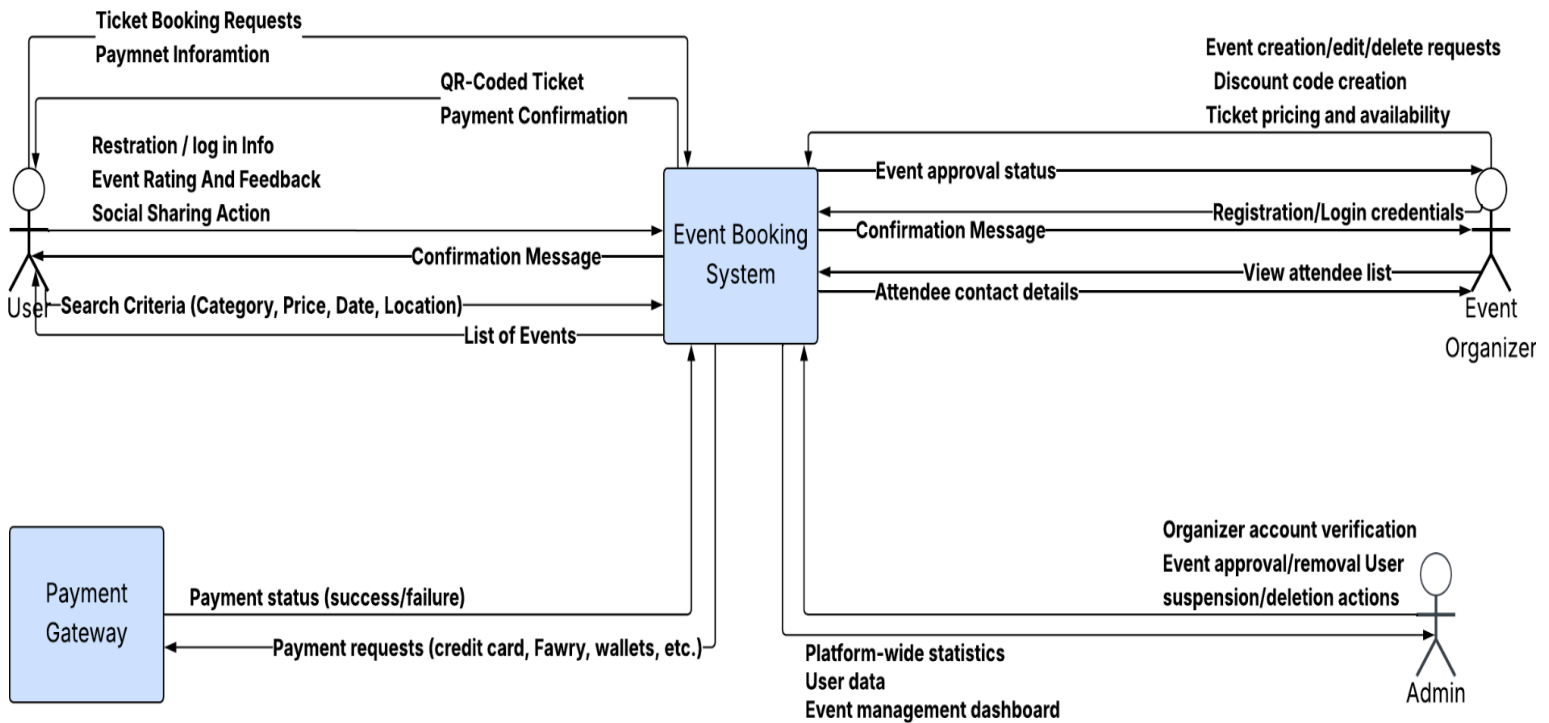
- Payment Gateways: Stripe, PayPal, and Razorpay.
- Database Systems: Firebase (for real-time sync) and PostgreSQL.
- Cloud Services: AWS or Azure for hosting, scaling, and CI/CD support.
- Calendar and Map APIs: Google Calendar and Google Maps for event reminders and location assistance.

4.4 Communications Interfaces

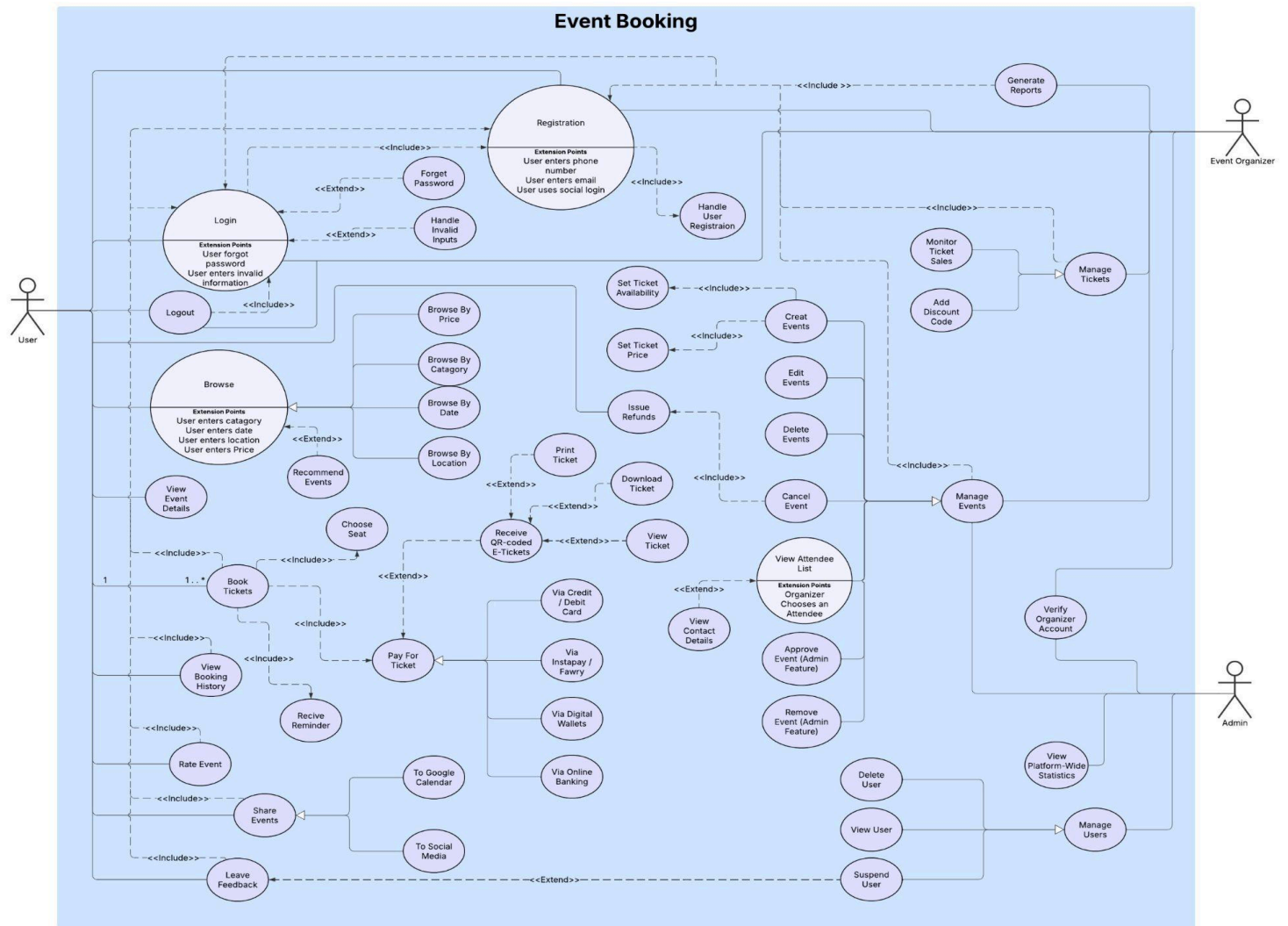
- Email/SMS Gateway: Used for confirmations, notifications, and reminders via providers such as SendGrid or Twilio.
 - Social Media Integration: Events can be shared via Facebook, Instagram, or WhatsApp.
 - Real-Time Updates: Firebase/WebSocket protocols for real-time seat selection and ticket availability.
 - Security Protocols: All communication over HTTPS using SSL encryption, with MFA enabled for sensitive operations.
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5. Organizing the Specific Requirements

5.1 Context Diagram



5.2 Complete use case model



5.2.2 Use case Narrative

1-Registration

Name:	Registration
ID:	UC01
Description:	A new user (User or an Event Organizer) registers into the system.
Actor:	System User Event Organizer
Trigger:	User clicks "Register" on the system's homepage.
Preconditions:	The user has access to the event booking platform. The user does not already have an account.
Postconditions:	A new user account is created and stored in the system. The user is ready to log in and use the platform.
Basic Flow:	1. User clicks "Register" on the homepage. 2. System displays the

	registration form with fields: - o Name - o Email - o Password - o Phone number (optional) - o Account type (User or Event Organizer) 3. User fills in details and selects account type. 4. User clicks "Submit". 5. System validates inputs: - o Email format is correct. - o Password meets complexity rules (e.g., 8+ chars, special characters). - o No duplicate email exists. 6. System creates the account and stores details in the database. 7. System sends a verification email . 8. System displays: "Registration successful ,Check your email to
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	verify.
Alternative Flow:	AF1: Social Media Registration : 1. User clicks "Register with Google/Facebook". 2. System redirects to the provider's login page. 3. User grants permission. 4. System retrieves: - o Email - o Name 5. System auto-fills the registration form (skipping password). 6. User selects account type (User or Event Organizer). 7. System completes registration without password. AF2: User registers with a phone number instead of email. 1. User selects "Register with Phone". 2. System prompts for: - o Phone number (mandatory) - o Name

	◦ Account type (User/Organizer) 3. User submits the form. 4. System sends an SMS OTP (One-Time Password) to the phone. 5. User enters the OTP. 6. System verifies the OTP and creates the account.
Exception Flow:	EF1: Duplicate Email: 1. System detects an existing email. 2. Displays error: "Email already registered. Try logging in or use another email." EF2: Weak Password: 1. Password does not meet complexity rules. 2. System shows: "Password must have 8+ characters, including a number and symbol." EF3: Social media login Failure: 1. Google/Facebook authentication fails. 2. System displays: "Social login failed. Try

	manual registration."
Special Requirements:	• Performance: Registration should complete in <3 seconds. • Security: SMS OTPs must be valid for 5 minutes only.

2-Login

Name:	Login
ID:	UC02
Description:	A registered user (User or Event Organizer) logs into their account.
Actor:	User Event Organizer
Trigger:	User clicks "Log in" on the system's homepage.
Preconditions:	User must have a registered account.
Postconditions:	Success: User is granted access to their dashboard. Failure: Error message is displayed, and login is retried.

Basic Flow:	1. User clicks "Log in" on the homepage. 2. System displays the login form with fields: - o Username/Email - o Password - o (Optional) "Remember Me" checkbox 3. User enters credentials and submits the form. 4. System validates: - o Username/email exists. - o Password matches the hashed value in the database. 5. If valid: - o System creates a session token (JWT or cookie). - o Redirects user to their dashboard/homepage.
Alternative Flow:	AF1: Forgot Password 1. User clicks "Forgot Password?". 2. System prompts for

	registered email/phone. 3. User submits email/phone. 4. System sends a password reset link (email) or OTP (SMS). 5. User resets password via the link/OTP. AF2: Social Login (OAuth) 1. User clicks "Log in with Google/Facebook". 2. OAuth provider authenticates the user. 3. System checks if the email exists in the database. - o If yes: Logs the user in. - o If no: Redirects to registration flow.
Exception Flow:	EF1: Invalid Credentials 1. User enters incorrect username/password. 2. System displays: <i>"Invalid username or password. Try again."</i> 3. System logs the failed attempt (for security monitoring). EF2: Account Locked

	1. User exceeds 3 failed login attempts. 2. System locks the account for 15 minutes. 3. Displays: <i>"Account locked. Try again later or reset your password."</i> EF3: Unverified Account 1. User's email/phone is unverified. 2. System displays: <i>"Please verify your email first." Resend verification link"</i>
Special Requirements:	• Performance: Registration should complete in <3 seconds. • Security: -SMS OTPs must be valid for 5 minutes only. -Passwords must be hashed.

3-Browse Events:

Name:	Browse events.
ID:	UC03

Description:	System User searches for events by price, filters, or categories.
Actor:	User
Trigger:	User clicks on the search bar or selects a filter/category from the event listing page
Preconditions:	User must be logged in (if required by business rules). The system must have active events in the database.
Postconditions:	Success: User sees a list of events matching their search/filters. Failure: System displays "No events found. Try different keywords/filters."
Basic Flow:	1. User clicks on the search bar or opens the events page. 2. User enters: ○ Event name (e.g., "Tech Conference 2024"). ○ Organizer name (e.g., "DevSummit Inc."). 3. System validates input (e.g., prevents SQL injection). 4. System queries the database and returns:

	○ Matching events (sorted by relevance/date). ○ Basic details: ■ Event name ■ Date & time ■ Location ■ Price range 5. User scrolls through results and clicks on an event for details.
Alternative Flow:	AF1: Filtered Search (Category/Price/Date/Location) 1. User clicks "Advanced Filters". 2. User selects: ○ Category (e.g., "Music," "Business"). ○ Price range (e.g., "Free," "50-50-100"). ○ Date range (e.g., "This weekend," "Next month"). ○ Location (e.g., "New York," "Online"). 3. System applies filters and displays refined results. AF2: Trending/Popular Events (Default View) 1. User lands on the events page without searching. 2. System displays:

	○ Trending events (based on user engagement). ○ Upcoming events (sorted by date).
Exception Flow:	EF1: No Results Found 1. User's search/filters return zero matches. 2. System displays: "No events found. Try adjusting your search or filters." ○ Suggestions (e.g., "Check spelling," "Browse popular events"). EF2: Server/Database Error 1. System fails to fetch events (e.g., API timeout). 2. Displays: "We couldn't load events. Please try again later."
Special Requirements:	UI: ● Autocomplete for event/organizer names. Performance: ● Search results load in < 1.5 seconds. ● Supports 10,000+ concurrent users.

4-Book a ticket:

Name:	Book Ticket
ID:	UC04
Description:	User wants to Book Tickets for an Event.
Actor:	User Secondary Actors: Payment Gateway (e.g., Stripe, PayPal) Email/SMS Service (for reminders)
Trigger:	User clicks "Book Tickets" on an event page.
Preconditions:	User is logged in (registered account required). Event is active (not canceled/sold out). User's account email/phone is verified.
Postconditions:	• Success: ○ Tickets are reserved in the system. ○ User receives a QR-code (email/app). ○ User gets reminders (2 days + 1 day before the event). • Failure:

	○ Payment fails , No ticket issued, error message shown. ○ Event sells out during booking ,User notified.
Basic Flow:	1. User clicks "Book Tickets" on the event page. 2. System checks: ○ Event availability (not sold out). ○ User's login status. 3. User selects: ○ Number of tickets (e.g., 2 adults, 1 child). ○ Seats (if reserved seating is enabled). 4. System calculates total cost (incl. fees/taxes). 5. User proceeds to checkout: ○ Selects payment method (credit card, PayPal, etc.). ○ Enters payment details. 6. Payment gateway processes the transaction: ○ If successful : System generates a QR-coded e-ticket. ○ Sends confirmation email/SMS with ticket and receipt.

	7. System schedules reminders: ○ 2 days before event: "Your event is coming up!" ○ 1 day before event: "Don't forget: Event tomorrow at [time]!"
Alternative Flow:	AF1: Save & Pay Later ● Scenario: User isn't ready to pay now. ● Steps: 1. User clicks "Hold Tickets." 2. System reserves tickets for 24 hours (no payment yet). 3. User returns later to complete payment.
Exception Flow:	EF1: Payment Failed 1. Credit card/PayPal transaction declines. 2. System displays: "Payment failed. Check your card details or try another method." 3. User retries or cancels booking. EF2: Event Sold Out 1. Tickets sell out during checkout.

	2. System shows: "Sorry, tickets are now sold out. Join the waitlist?"
Special Requirements:	Security: • Payment data never stored (PCI-DSS compliant). • QR codes encrypted (to prevent forgery). Performance: • Seat selection updates in real-time (no double bookings). • Payment processing <5 seconds. UI: • Auto-refresh for seat availability. • Progress indicator during checkout (e.g., "Step 2/4").

Appendix A: A.1 Data Dictionary

A.1.1 Introduction

This data dictionary documents all data entities, attributes, relationships, and constraints for the Event Booking System.

A.1.2 Tables & Field

Users:

Field Name	Data Type	Description	Constraints
user_id	ID	Unique identifier for users	PRIMARY KEY, NOT NULL
email	VARCHAR	User's email address	UNIQUE, NOT NULL, max 100 chars
password	VARCHAR	Encrypted password	NOT NULL, min 8 chars
first_name	VARCHAR	User's first name	max 50 chars
last_name	VARCHAR	User's last name	max 50 chars
phone	VARCHAR	Contact number	Optional, format validated
role	ENUM	User role (attendee/organizer/admin)	NOT NULL, values: 'attendee', 'organizer', 'admin'
created_at	DATE	Account creation date	NOT NULL

2.Event:

Field Name	Data Type	Description	Constraints
event_id	ID	Unique event identifier	PRIMARY KEY, NOT NULL
organizer_id	ID	ID of organizing user	FOREIGN KEY (Users.user_id), NOT NULL
user_id	ID	ID of attending user	FOREIGN KEY (Users.user_id), NOT NULL

title	VARCHAR	Event title	NOT NULL, max 200 chars
description	TEXT	Detailed event description	Optional
start_date	DATETIME	Event start date/time	NOT NULL
end_date	DATETIME	Event end date/time	Optional
venue_id	ID	Event location	FOREIGN KEY (Venues.venue_id)
max_attendees	INTEGER	Maximum ticket capacity	min: 1
status	ENUM	Event status	values: 'draft', 'active', 'cancelled', 'completed'

3. Venue

Field Name	Data Type	Description	Constraints
venue_id	ID	Unique venue identifier	PRIMARY KEY, NOT NULL
name	VARCHAR	Venue name	NOT NULL, max 100 chars
address	TEXT	Full physical address	NOT NULL
capacity	INTEGER	Maximum occupancy	min: 1
seating_map_url	VARCHAR	URL to seating layout image	Optional

4. Booking

Field Name	Data Type	Description	Constraints
booking_id	ID	Unique booking identifier	PRIMARY KEY, NOT NULL

user_id	ID	Unique identifier for users	FOREIGN KEY (Users.user_id),, NOT NULL
event_id	ID	Unique event identifier	FOREIGN KEY (Event.event_id), NOT NULL
ticket_count	INTEGER	Number of tickets booked	min: 1
booking_date	DATE	When booking was made	NOT NULL
status	ENUM	Event status	values: 'confirmed', 'cancelled', 'refunded'

6. Payment

Field Name	Data Type	Description	Constraints
payment_id	ID	Unique payment identifier	PRIMARY KEY, NOT NULL
booking_id	ID	Unique booking identifier	FOREIGN KEY (Booking.booking_id),, NOT NULL
amount	DECIMAL	Payment amount	NOT NULL, precision: 10, scale: 2
payment_method	ENUM	Payment processor used	values: 'credit_card', 'fowary', 'bank_transfer'
status	ENUM	Payment status	values: 'pending', 'completed', 'failed', 'refunded'

A.1.3 Relationships

Relationship	Description	Implementation
User → Events	One organizer can create many events	Events.organizer_id → Users.user_id

Event → Bookings	One event can have many bookings	Bookings.event_id → Events.event_id
User → Bookings	One user can have many bookings	Bookings.user_id → Users.user_id
Booking → Payment	One booking has one payment	Payments.booking_id → Bookings.booking_id
Event → Venue	One event occurs at one venue	Events.venue_id → Venues.venue_id

A.1.4 Enumerations

Booking status

Value	Description
confirmed	Booking is active
cancelled	User cancelled booking
refunded	Payment was refunded

Payment status

Value	Description
pending	Payment initiated
completed	Successfully processed
failed	Payment declined

refunded	Amount returned to user
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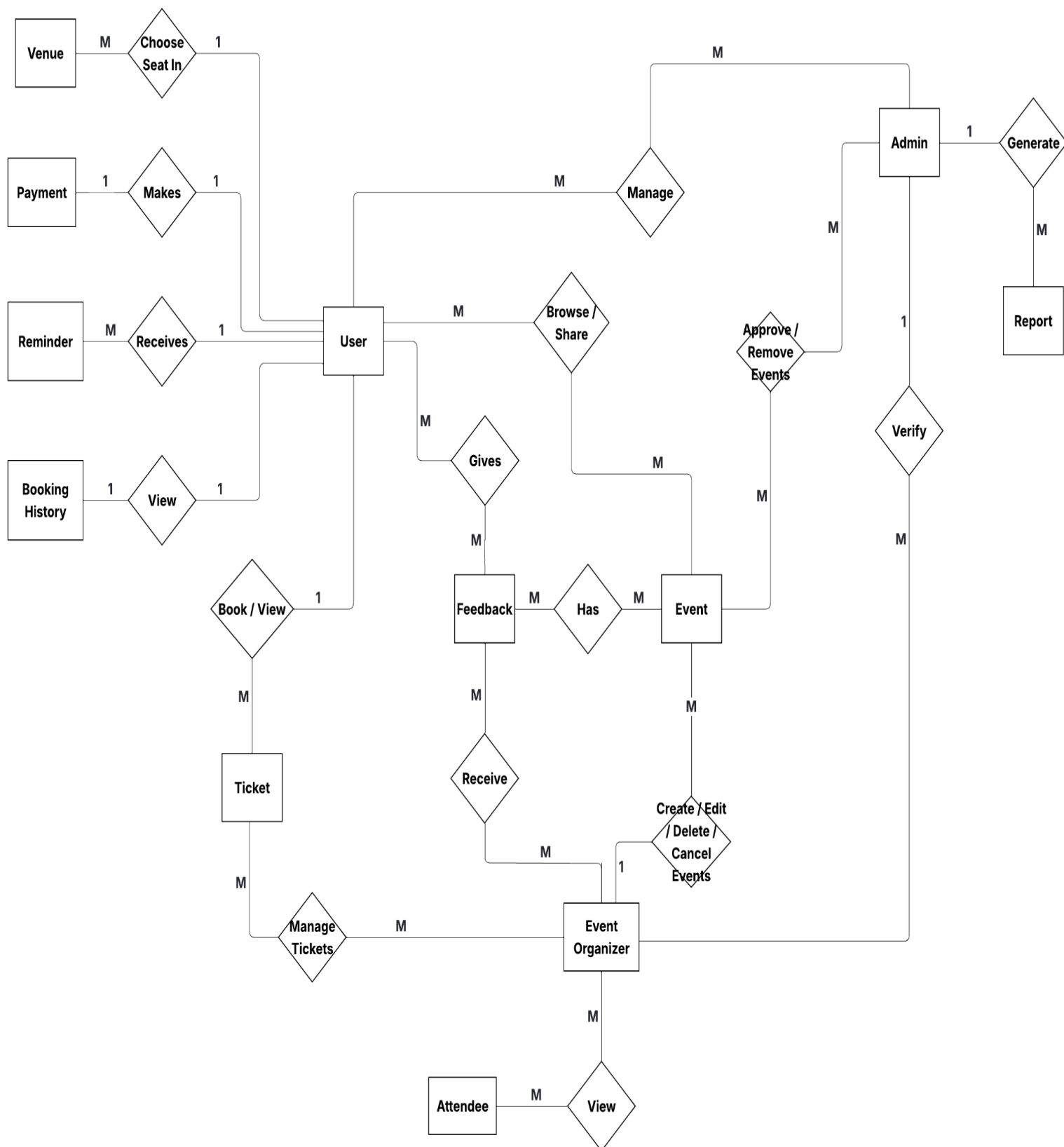
User Role

Value	Description
attendee	Regular event attendee
organizer	Event creator/manager

Event status

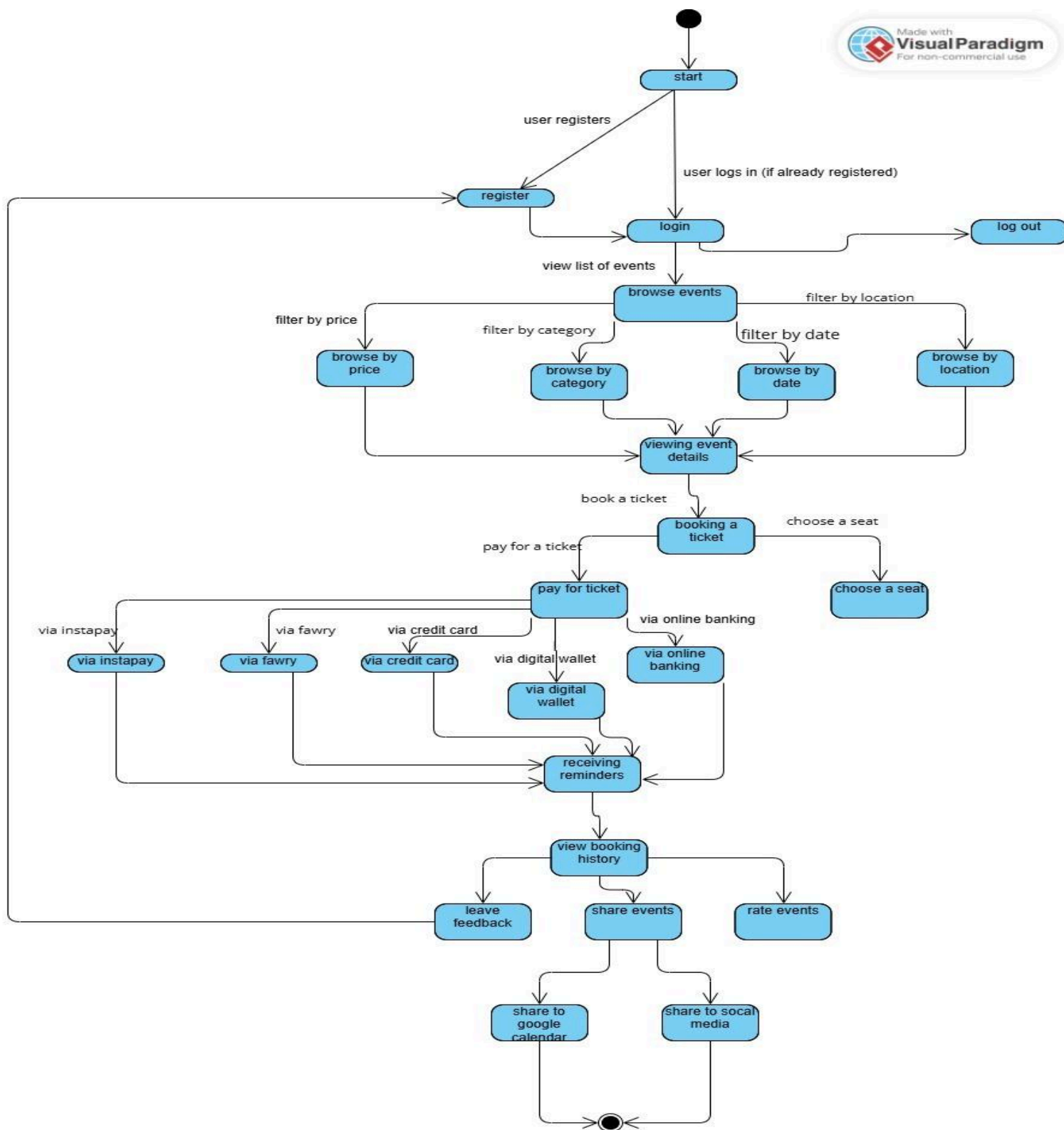
Value	Description
draft	Unpublished event
active	Available for booking
cancelled	Event cancelled
completed	Event has concluded

A.2 Data Model :

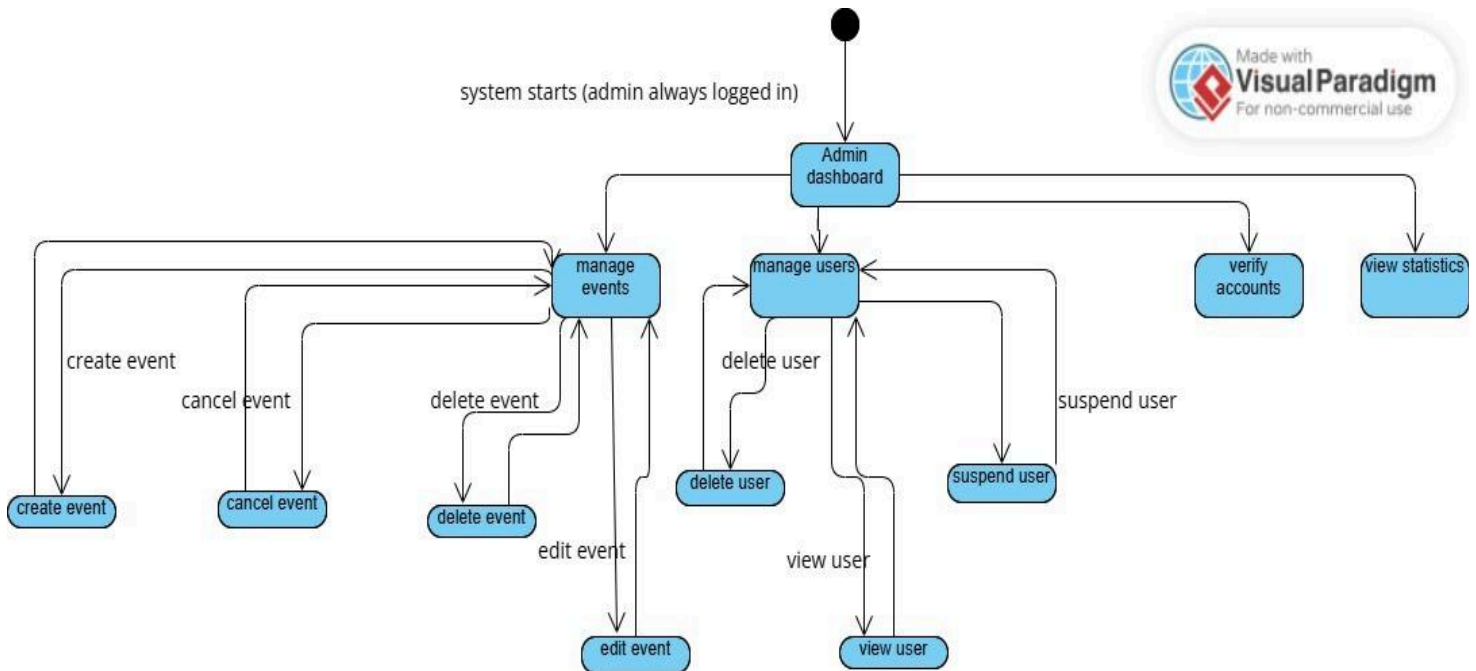


Appendix B :

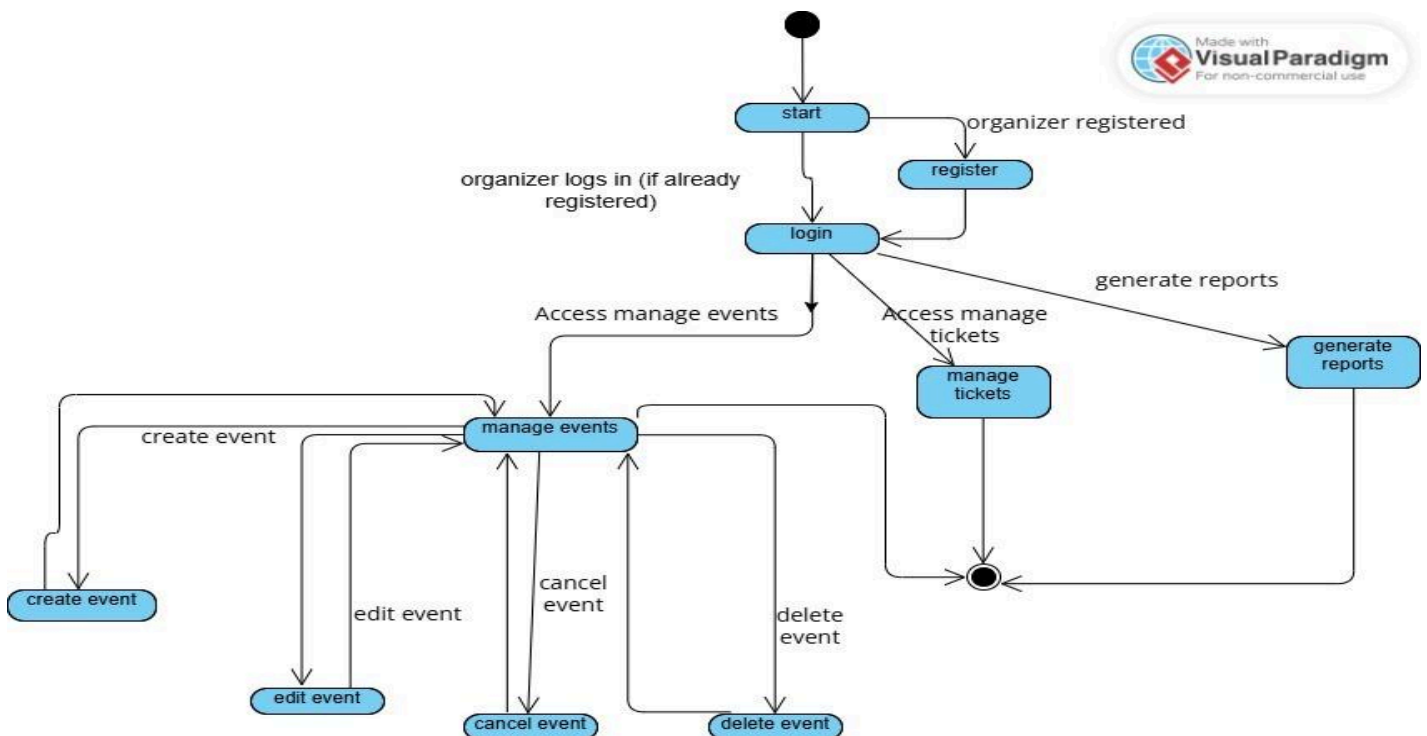
User State - Transition Diagram



Admin State - Transition Diagram



Event Organizer State - Transition Diagram



6. Use case table

1. Registration

Name	Registration
ID	EB11
Description	User wants to Register into the system
Actors	System Users, Event Organizers
Preconditions	-
Postconditions	The User Registers into the System successfully
Main course	1. User clicks Register 2. User fills the required information (Name, password, phone number) 3. User Chooses weather it is a User/Event Organizer Account 4. User submits Registration form 5. The System Registers the User successfully
Alternative course	The User Registers with their google account or other social media (facebook, instagram, etc ...)
Exception	-
Name	Registration
course	

2. Login

Name	Login
ID	EB12
Description	User wants to log in into account

Actors	System Users, Event Organizers
Preconditions	The User must have a Registered account in the System
Postconditions	If Username and Password are correct -> the user is granted access to the system If they are incorrect -> an error message is displayed and the user gets to try Logging in again
Main course	1. User clicks Log in 2. User enters Username and Password 3. User submits Login form 4. the user is granted access to the system
Alternative course	If the user forgets his password, they can use the forget password option to reset it
Exception course	Error message is displayed and the user gets to try Logging in again if the Username and Password are incorrect

3. Logout

Name	Logout
ID	EB13
Description	User wants to Logout
Name	Logout
Actors	System Users
Preconditions	The User must have a Registered account in the System and must be Logged in
Postconditions	The User is Logged out successfully and loses access to the System

Main course	1. User clicks Logout 2. The System Logs out the User successfully
Alternative course	-
Exception course	-

4. Browse

Name	Browse
ID	EB14
Description	User wants to look for Events
Actors	System Users
Preconditions	-
Postconditions	The User gets results for the event they are looking for
Main course	1. User clicks the Search Bar 2. User enters event Name/ Event Organizer's Name 3. User gets the results for their search
Alternative course	The User tries searching by Category, Price, Date or Location
Name	Browse
Exception course	The Event the User is looking for is not in the system so a message appears saying " Results unavailable "

5. View Event Details

Name	View Event Details
ID	EB15
Description	User wants to see Event Details
Actors	System Users
Preconditions	-
Postconditions	The User gets to see Event Details
Main course	1. User clicks on the Event 2. A page loads with all the Event Details
Alternative course	-
Exception course	-

6. Book Tickets

Name	Book Tickets
ID	EB16
Description	User wants to Book Tickets for an Event
Actors	System Users
Preconditions	The User must have a Registered account in the System and must be Logged in
Name	Book Tickets
Postconditions	The User Chooses the number of Tickets, seats, pays for the Tickets and receive reminders for the event

Main course	1. User clicks on an event 2. User clicks on Book Tickets 3. User Chooses number of Tickets and the seats 4. User Pays for the Tickets 5. User receives a QR-Coded Ticket 6. User receives a Reminder 2 days and 1 day before the date of the event
Alternative course	-
Exception course	If payment is unsuccessful the user doesn't receive a QR-Coded Ticket and gets an error message instead

7. View Booking History

Name	View Booking History
ID	EB17
Description	User wants to View Booking History
Actors	System Users
Preconditions	The User must have a Registered account in the System and must be Logged in
Postconditions	The User sees the Booking History
Main course	1. User clicks on View Booking History 2. User sees Booking History
Alternative course	-
Exception	If the User didn't Book anything before then the Page is
Name	View Booking History

course	empty with the message “ History is empty ”
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8. Rate Events

Name	Rate Events
ID	EB18
Description	User wants to Rate Events
Actors	System Users
Preconditions	The User must have a Registered account in the System and must be Logged in
Postconditions	The User Chooses the rating for the event and submits it successfully
Main course	1. User clicks on the Event 2. User clicks on Rate Event 3. User Chooses the Rating (1 - 5 stars) 4. User submits the Rating
Alternative course	-
Exception course	-

9. Share Events

Name	Share Events
ID	EB19
Description	User wants to Share Events on other social media
Name	Share Events

Actors	System Users
Preconditions	The User must have a Registered account in the System and must be Logged in
Postconditions	The User Shares the Event successfully
Main course	1. User clicks on Event 2. User clicks on Share Event 3. User Chooses the social they want to share to
Alternative course	-
Exception course	-

10. Leave Feedback

Name	Leave Feedback
ID	EB110
Description	User wants to Leave FeedBack on an event
Actors	System Users
Preconditions	The User must have a Registered account in the System and must be Logged in
Postconditions	The User write Feedback on the event and submits successfully
Main course	1. User clicks on the Event 2. User clicks on the Comment button 3. User writes a Comment 4. User submits the Comment
Name	Leave Feedback

Alternative course	-
Exception course	If the user writes a Comment against the community guidelines the Comment is deleted and the User receives a warning, after three warnings the Users account is Suspended

11. Generate Reports

Name	Generate Reports
ID	EB21
Description	The Event Organizer Wants to Generate Reports based on Ticket sales, Revenue, and Attendance statistics.
Actors	Event Organizers
Preconditions	The Event Organizers must have a Registered account in the System and must be Logged in
Postconditions	The Event Organizer Generates Reports successfully
Main course	1. The Event Organizer Clicks on Generate Report 2. The Event Organizer Chooses what the Report is based on 3. The report is generated Successfully
Alternative course	-
Exception course	-

12. Manage Events

Name	Manage Events
ID	EB22
Description	The Event Organizer and Admin want to Manage Events
Actors	Event Organizer, Admins
Preconditions	The Event Organizer must have a Registered account in the System and must be Logged in, Admins are always Logged in by default
Postconditions	- The Event Organizer Creates/ Deletes/ Edits/ Cancels Events successfully - Admin Approves/ Removes Events successfully
Main course	1. The Event Organizer/ Admin clicks on Manage Events 2. The Event Organizer/ Admin Chooses what they want to do 3. The Events are edited successfully
Alternative course	-
Exception course	-

13. Manage Tickets

Name	Manage Tickets
ID	EB23
Description	The Event Organizer wants to Manage Tickets
Actors	Event Organizer

Preconditions	The Event Organizer must have a Registered account in the System and must be Logged in
Name	Manage Tickets
Postconditions	The Event Organizer Sets the Availability/ Sets Price successfully
Main course	1. The Event Organizer clicks on Manage Tickets 2. The Event Organizer Chooses what they want to do (Monitor Ticket Sales or Add Discount Code to Tickets)
Alternative course	-
Exception course	-

14. Verify Organizer Account

Name	Verify Organizer Account
ID	EB31
Description	The Admin wants to verify Event Organizer's Accounts
Actors	Admins
Preconditions	-
Postconditions	The Admin Verifies the Event Organizer's Account successfully
Main course	1. Admin clicks on the Event Organizer's Account 2. Admin clicks on Verify Account 3. Account is Verified successfully

Alternative course	-
Exception course	-

15. View Platform-Wide Statistics

Name	View Platform-Wide Statistics
ID	EB32
Description	The Admin wants to View Platform-Wide Statistics
Actors	Admins
Preconditions	-
Postconditions	The Admin sees the Platform-Wide Statistics successfully
Main course	1. Admin clicks on View Platform-Wide Statistics 2. A page loads with Platform-Wide Statistics successfully
Alternative course	-
Exception course	-

16. Manage Users

Name	Manage Users
ID	EB33
Description	The Admin wants to View/ Suspend/ Delete Users
Actors	Admins, Users

Preconditions	-
Postconditions	The Admin edits Users successfully
Main course	1. Admin clicks on a User Account 2. Admin clicks on Manage Account 3. Admin chooses what they want to do (View/
Name	Manage Users
	Suspend/ Delete Users) 4. Users are edited successfully
Alternative course	-
Exception course	-